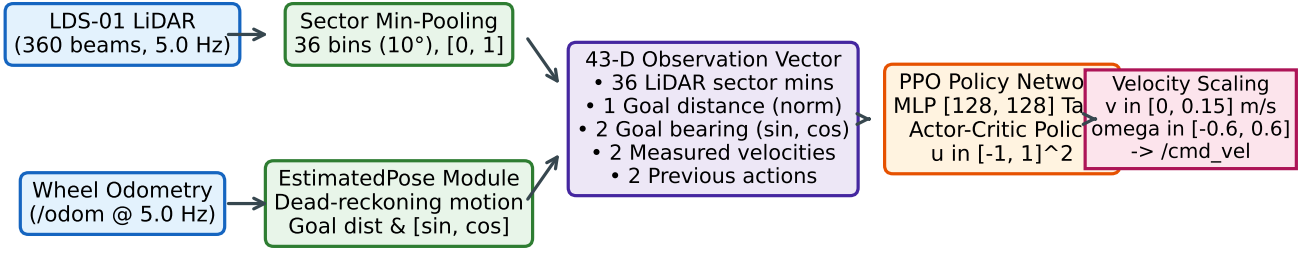
LDS-01 LiDAR
(360 beams, 5.0 Hz)
Sector Min-Pooling
36 bins (10°), [0, 1]
Wheel Odometry
(/odom @ 5.0 Hz)
EstimatedPose Module
Dead-reckoning motion
Goal dist & [sin, cos]
43-D Observation Vector
• 36 LiDAR sector mins
• 1 Goal distance (norm)
• 2 Goal bearing (sin, cos)
• 2 Measured velocities
• 2 Previous actions
PPO Policy Netwo
MLP [128, 128] Ta
Actor-Critic Polic
u in [-1, 1]^2
Velocity Scaling
v in [0, 0.15] m/s
omega in [-0.6, 0.6]
-> /cmd_vel